Experiment-10

View Queries-

1. Create a view with name ‘v1’ using employees1 table which holds the value of employee_id and salary of employee.
2. Do the insert and delete records from v1 table.

```
mysql> select*from Employee;
+------+----------+----------+------------+--------+
| E_id | F_Name   | L_Name   | dev_serial | salary |
+------+----------+----------+------------+--------+
|    1 | Arun     | Kumar    | DEV001     |  25000 |
|    2 | Priya    | Sharma   | DEV002     |  32000 |
|    3 | Charan   | Raj      | DEV003     |  28000 |
|    4 | Divya    | Lakshmi  | DEV004     |  35000 |
|    5 | Kavin    | Kumar    | DEV005     |  30000 |
|    6 | Anitha   | Devi     | DEV006     |  40000 |
|    7 | Rahul    | Verma    | DEV007     |  27000 |
|    8 | Sneha    | Reddy    | DEV008     |  45000 |
|    9 | Bharath  | Raj      | DEV009     |  38000 |
|   10 | Meena    | Kumari   | DEV010     |  29000 |
|   11 | Vijay    | Anand    | DEV011     |  50000 |
|   12 | Karthik  | Rajan    | DEV012     |  42000 |
|   13 | Nisha    | Priya    | DEV013     |  31000 |
|   14 | Santhosh | Kumar    | DEV014     |  36000 |
|   15 | Deepa    | Rani     | DEV015     |  33000 |
|   16 | Ajay     | Krishnan | DEV016     |  48000 |
|   17 | Harini   | Devi     | DEV017     |  26000 |
|   18 | Manoj    | Kumar    | DEV018     |  39000 |
|   19 | Keerthi  | Rajan    | DEV019     |  34000 |
|   20 | Suresh   | Babu     | DEV020     |  55000 |
+------+----------+----------+------------+--------+
20 rows in set (0.00 sec)
```

```
mysql> desc Employee;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| E_id       | int         | YES  |     | NULL    |       |
| F_Name     | varchar(50) | YES  |     | NULL    |       |
| L_Name     | varchar(50) | YES  |     | NULL    |       |
| dev_serial | varchar(15) | YES  |     | NULL    |       |
| salary     | int         | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
mysql> create view v1 as select E_id, salary from Employee;
Query OK, 0 rows affected (0.00 sec)

mysql> desc v1;
+--------+------+------+-----+---------+-------+
| Field  | Type | Null | Key | Default | Extra |
+--------+------+------+-----+---------+-------+
| E_id   | int  | YES  |     | NULL    |       |
| salary | int  | YES  |     | NULL    |       |
+--------+------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

```
mysql> insert into v1 values(101,1000001);
Query OK, 1 row affected (0.00 sec)

mysql> select*from v1;
+------+---------+
| E_id | salary  |
+------+---------+
|    1 |   25000 |
|    2 |   32000 |
|    3 |   28000 |
|    4 |   35000 |
|    5 |   30000 |
|    6 |   40000 |
|    7 |   27000 |
|    8 |   45000 |
|    9 |   38000 |
|   10 |   29000 |
|   11 |   50000 |
|   12 |   42000 |
|   13 |   31000 |
|   14 |   36000 |
|   15 |   33000 |
|   16 |   48000 |
|   17 |   26000 |
|   18 |   39000 |
|   19 |   34000 |
|   20 |   55000 |
|  101 | 1000001 |
+------+---------+
21 rows in set (0.00 sec)
```

```
mysql> select*from Employee;
+------+----------+----------+------------+---------+
| E_id | F_Name   | L_Name   | dev_serial | salary  |
+------+----------+----------+------------+---------+
|    1 | Arun     | Kumar    | DEV001     |   25000 |
|    2 | Priya    | Sharma   | DEV002     |   32000 |
|    3 | Charan   | Raj      | DEV003     |   28000 |
|    4 | Divya    | Lakshmi  | DEV004     |   35000 |
|    5 | Kavin    | Kumar    | DEV005     |   30000 |
|    6 | Anitha   | Devi     | DEV006     |   40000 |
|    7 | Rahul    | Verma    | DEV007     |   27000 |
|    8 | Sneha    | Reddy    | DEV008     |   45000 |
|    9 | Bharath  | Raj      | DEV009     |   38000 |
|   10 | Meena    | Kumari   | DEV010     |   29000 |
|   11 | Vijay    | Anand    | DEV011     |   50000 |
|   12 | Karthik  | Rajan    | DEV012     |   42000 |
|   13 | Nisha    | Priya    | DEV013     |   31000 |
|   14 | Santhosh | Kumar    | DEV014     |   36000 |
|   15 | Deepa    | Rani     | DEV015     |   33000 |
|   16 | Ajay     | Krishnan | DEV016     |   48000 |
|   17 | Harini   | Devi     | DEV017     |   26000 |
|   18 | Manoj    | Kumar    | DEV018     |   39000 |
|   19 | Keerthi  | Rajan    | DEV019     |   34000 |
|   20 | Suresh   | Babu     | DEV020     |   55000 |
|  101 | NULL     | NULL     | NULL       | 1000001 |
+------+----------+----------+------------+---------+
21 rows in set (0.00 sec)
```

```
mysql> delete from v1 where E_id=10;
Query OK, 1 row affected (0.00 sec)

mysql> select*from v1;
+------+---------+
| E_id | salary  |
+------+---------+
|    1 |   25000 |
|    2 |   32000 |
|    3 |   28000 |
|    4 |   35000 |
|    5 |   30000 |
|    6 |   40000 |
|    7 |   27000 |
|    8 |   45000 |
|    9 |   38000 |
|   11 |   50000 |
|   12 |   42000 |
|   13 |   31000 |
|   14 |   36000 |
|   15 |   33000 |
|   16 |   48000 |
|   17 |   26000 |
|   18 |   39000 |
|   19 |   34000 |
|   20 |   55000 |
|  101 | 1000001 |
+------+---------+
20 rows in set (0.00 sec)
```

```
mysql> select*from Employee;
+------+----------+----------+------------+---------+
| E_id | F_Name   | L_Name   | dev_serial | salary  |
+------+----------+----------+------------+---------+
|    1 | Arun     | Kumar    | DEV001     |   25000 |
|    2 | Priya    | Sharma   | DEV002     |   32000 |
|    3 | Charan   | Raj      | DEV003     |   28000 |
|    4 | Divya    | Lakshmi  | DEV004     |   35000 |
|    5 | Kavin    | Kumar    | DEV005     |   30000 |
|    6 | Anitha   | Devi     | DEV006     |   40000 |
|    7 | Rahul    | Verma    | DEV007     |   27000 |
|    8 | Sneha    | Reddy    | DEV008     |   45000 |
|    9 | Bharath  | Raj      | DEV009     |   38000 |
|   11 | Vijay    | Anand    | DEV011     |   50000 |
|   12 | Karthik  | Rajan    | DEV012     |   42000 |
|   13 | Nisha    | Priya    | DEV013     |   31000 |
|   14 | Santhosh | Kumar    | DEV014     |   36000 |
|   15 | Deepa    | Rani     | DEV015     |   33000 |
|   16 | Ajay     | Krishnan | DEV016     |   48000 |
|   17 | Harini   | Devi     | DEV017     |   26000 |
|   18 | Manoj    | Kumar    | DEV018     |   39000 |
|   19 | Keerthi  | Rajan    | DEV019     |   34000 |
|   20 | Suresh   | Babu     | DEV020     |   55000 |
|  101 | NULL     | NULL     | NULL       | 1000001 |
+------+----------+----------+------------+---------+
20 rows in set (0.00 sec)
```

Index queries:

1. Create index1 for ‘salary’ attribute from employees1 relation and list the first name of the employees whose salary is above 145000 and explain the working principle of indexing and then drop the index1.
2. Create index1 for ‘employee_id’ attribute and display the first name of an employee whose employee id is 10 and explain the working principle of index1.

```
mysql> select*from Employee;
+------+----------+----------+------------+--------+
| E_id | F_Name   | L_Name   | dev_serial | salary |
+------+----------+----------+------------+--------+
|    1 | Arun     | Kumar    | DEV001     | 125000 |
|    2 | Priya    | Sharma   | DEV002     | 152000 |
|    3 | Charan   | Raj      | DEV003     | 118000 |
|    4 | Divya    | Lakshmi  | DEV004     | 135000 |
|    5 | Kavin    | Kumar    | DEV005     | 130000 |
|    6 | Anitha   | Devi     | DEV006     | 160000 |
|    7 | Rahul    | Verma    | DEV007     | 127000 |
|    8 | Sneha    | Reddy    | DEV008     | 145000 |
|    9 | Bharath  | Raj      | DEV009     | 138000 |
|   10 | Meena    | Kumari   | DEV010     | 119000 |
|   11 | Vijay    | Anand    | DEV011     | 140000 |
|   12 | Karthik  | Rajan    | DEV012     | 142000 |
|   13 | Nisha    | Priya    | DEV013     | 131000 |
|   14 | Santhosh | Kumar    | DEV014     | 136000 |
|   15 | Deepa    | Rani     | DEV015     | 133000 |
|   16 | Ajay     | Krishnan | DEV016     | 144000 |
|   17 | Harini   | Devi     | DEV017     | 116000 |
|   18 | Manoj    | Kumar    | DEV018     | 139000 |
|   19 | Keerthi  | Rajan    | DEV019     | 134000 |
|   20 | Suresh   | Babu     | DEV020     | 145000 |
+------+----------+----------+------------+--------+
20 rows in set (0.00 sec)
```

```
mysql> select F_Name from Employee Where salary>145000;
+---------+
| F_Name  |
+---------+
| Priya   |
| Anitha  |
+---------+
2 rows in set (0.00 sec)
```

```
mysql> Explain select F_Name from Employee Where salary>145000;
+----+-------------+----------+------------+------+---------------+------+---------+------+------+----------+-------------+
| id | select_type | table    | partitions | type | possible_keys | key  | key_len | ref  | rows | filtered | Extra       |
+----+-------------+----------+------------+------+---------------+------+---------+------+------+----------+-------------+
|  1 | SIMPLE      | Employee | NULL       | ALL  | NULL          | NULL | NULL    | NULL |   20 |    33.33 | Using where |
+----+-------------+----------+------------+------+---------------+------+---------+------+------+----------+-------------+
1 row in set, 1 warning (0.00 sec)

mysql> create index index1 on Employee(salary);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> Explain select F_Name from Employee Where salary>145000;
+----+-------------+----------+------------+-------+---------------+--------+---------+------+------+----------+-----------------------+
| id | select_type | table    | partitions | type  | possible_keys | key    | key_len | ref  | rows | filtered | Extra                 |
+----+-------------+----------+------------+-------+---------------+--------+---------+------+------+----------+-----------------------+
|  1 | SIMPLE      | Employee | NULL       | range | index1        | index1 | 5       | NULL |    2 |   100.00 | Using index condition |
+----+-------------+----------+------------+-------+---------------+--------+---------+------+------+----------+-----------------------+
1 row in set, 1 warning (0.00 sec)
```

```
mysql> drop index index1 on Employee;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> select F_Name from Employee Where E_id=10;
+---------+
| F_Name |
+---------+
| Meena  |
+---------+
1 row in set (0.00 sec)

mysql> Explain select F_Name from Employee Where E_id=10;
+----+-------------+----------+------------+------+---------------+------+---------+------+------+----------+-------------+
| id | select_type | table    | partitions | type | possible_keys | key  | key_len | ref  | rows | filtered | Extra       |
+----+-------------+----------+------------+------+---------------+------+---------+------+------+----------+-------------+
|  1 | SIMPLE      | Employee | NULL       | ALL  | NULL          | NULL | NULL    | NULL |   20 |    10.00 | Using where |
+----+-------------+----------+------------+------+---------------+------+---------+------+------+----------+-------------+
1 row in set, 1 warning (0.00 sec)

mysql> create index index1 on Employee(E_id);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> Explain select F_Name from Employee Where E_id=10;
+----+-------------+----------+------------+------+---------------+--------+---------+-------+------+----------+-------+
| id | select_type | table    | partitions | type | possible_keys | key    | key_len | ref   | rows | filtered | Extra |
+----+-------------+----------+------------+------+---------------+--------+---------+-------+------+----------+-------+
|  1 | SIMPLE      | Employee | NULL       | ref  | index1        | index1 | 5       | const |    1 |   100.00 | NULL  |
+----+-------------+----------+------------+------+---------------+--------+---------+-------+------+----------+-------+
1 row in set, 1 warning (0.00 sec)
```

```
mysql> show index from Employee;
+----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table    | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expressior |
+----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Employee |          1 | PRIMARY  |            1 | E_id        | A         |          20 |     NULL |   NULL |      | BTREE      |         |               |   YES   |  NULL      |
+----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
1 row in set (0.00 sec)
```